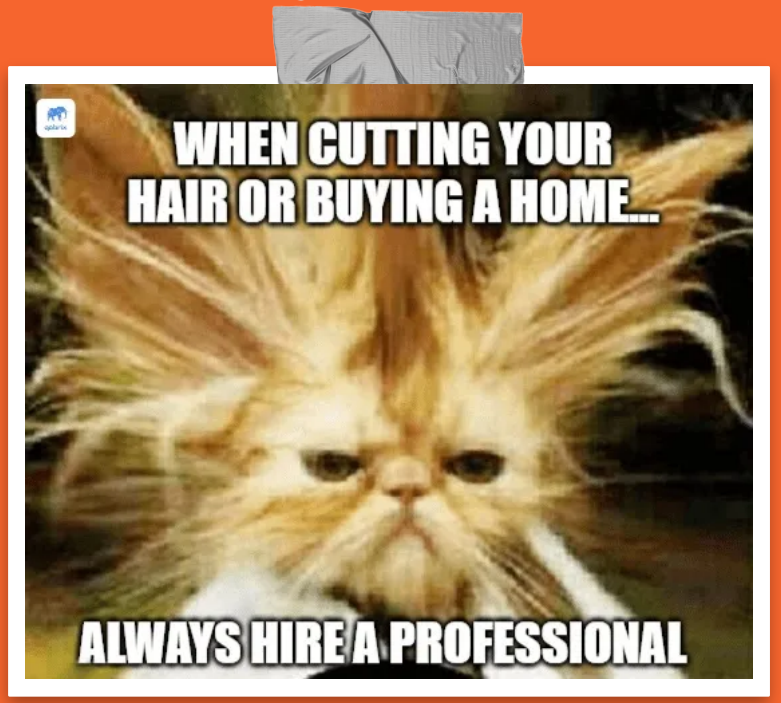
King County Housing Data: Where Nicole Should Buy ?



Insights for a mid-priced home in a lively, central neighborhood

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Neuefische Bootcamp - Data Science and AI

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The Client

Mid-priced home

Her budget sits in the middle of the market: $322,000–$645,000 (the middle 50% of all prices).

Lively and central

She wants to live close to downtown Seattle, in an active neighborhood.

Within the next year

She plans to buy within a year, so the time of purchase is her choice.



NICOLE JOHNSON

BUYER

- MID-PRICED HOME
- LIVELY & CENTRAL
- BUY WITHIN A YEAR

What is the KCHD?

Dataset: One year of house sales in King County

21,597
house sales

12 months
of sales, May 2014 – May 2015

Sale information

- price
- date of sale
- waterfront
- quality of view
- overall condition (1–5)
- King County grade (1–13)

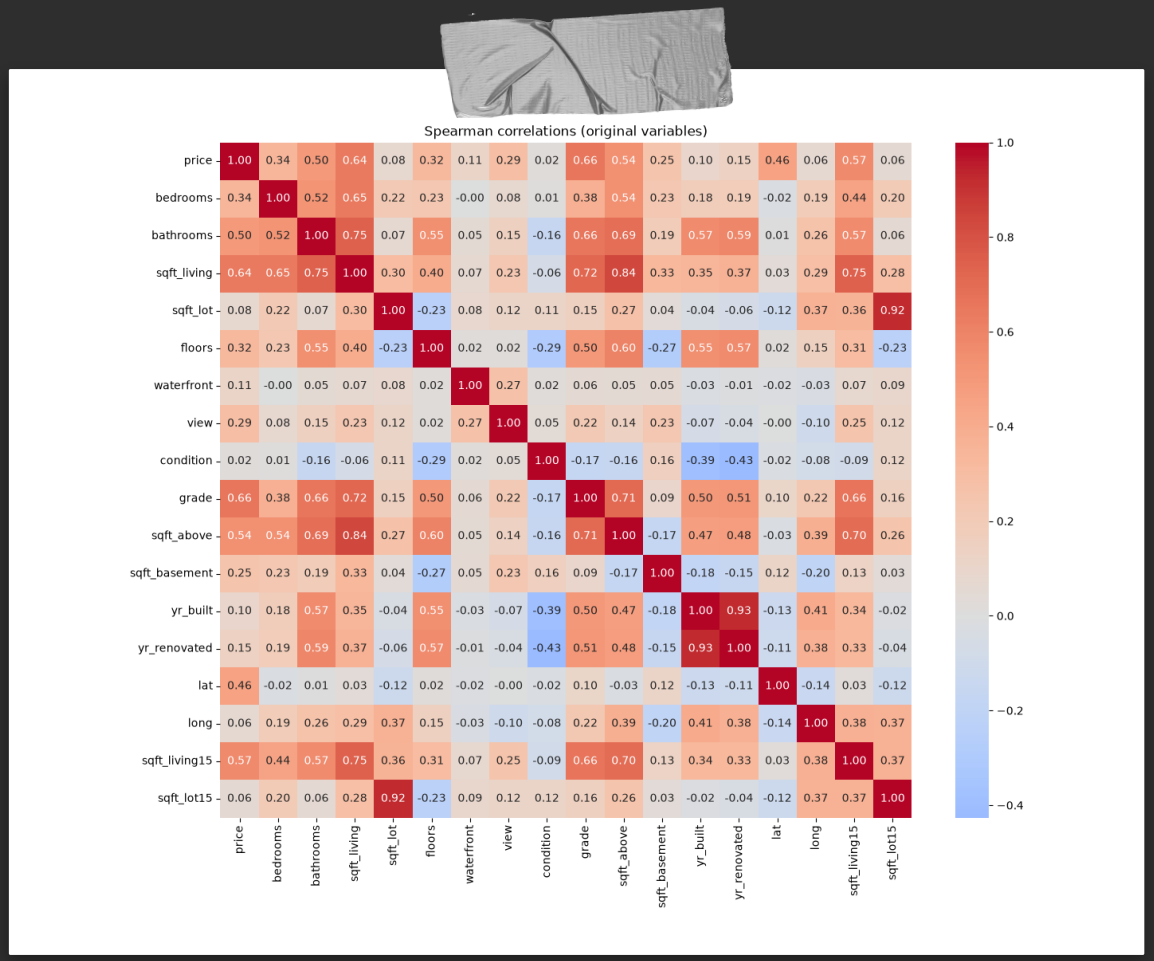
About the building

- bedrooms and bathrooms
- living area (sq ft)
- lot size (sq ft)
- floors and basement
- year built
- year renovated

The location

- zip code
- latitude and longitude
- living area of the 15 nearest neighbors
- lot size of the 15 nearest neighbors

What drives the price?



Size and build quality matter most.

Grade (0.66) and living area (0.64) have the strongest link with price.

Location matters too.

Houses further north are more expensive (latitude 0.46).

Age and condition barely matter on their own.

Year built (0.10) and condition (0.02) show almost no link with price.

Spearman correlation: 0 = no link, 1 = perfect link

1. Size matters more than the number of rooms

Price rises with living area

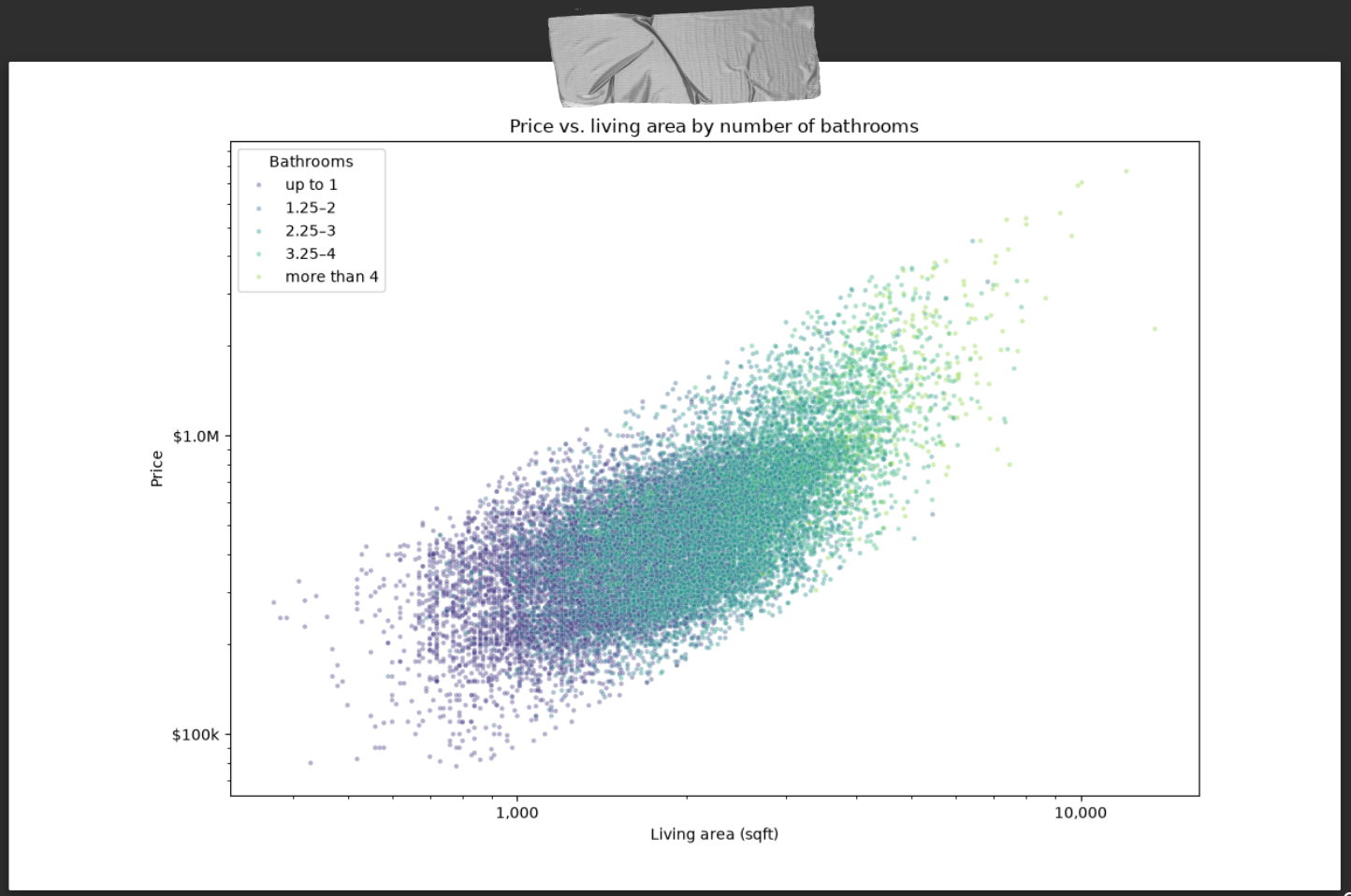
Bigger houses cost more, almost in a straight line.

Bathrooms come with size

Houses with more bathrooms sit further right, because extra bathrooms mostly come with bigger houses.

Little extra value

At the same size, the groups overlap: bathrooms (0.50) add little beyond living area (0.64).



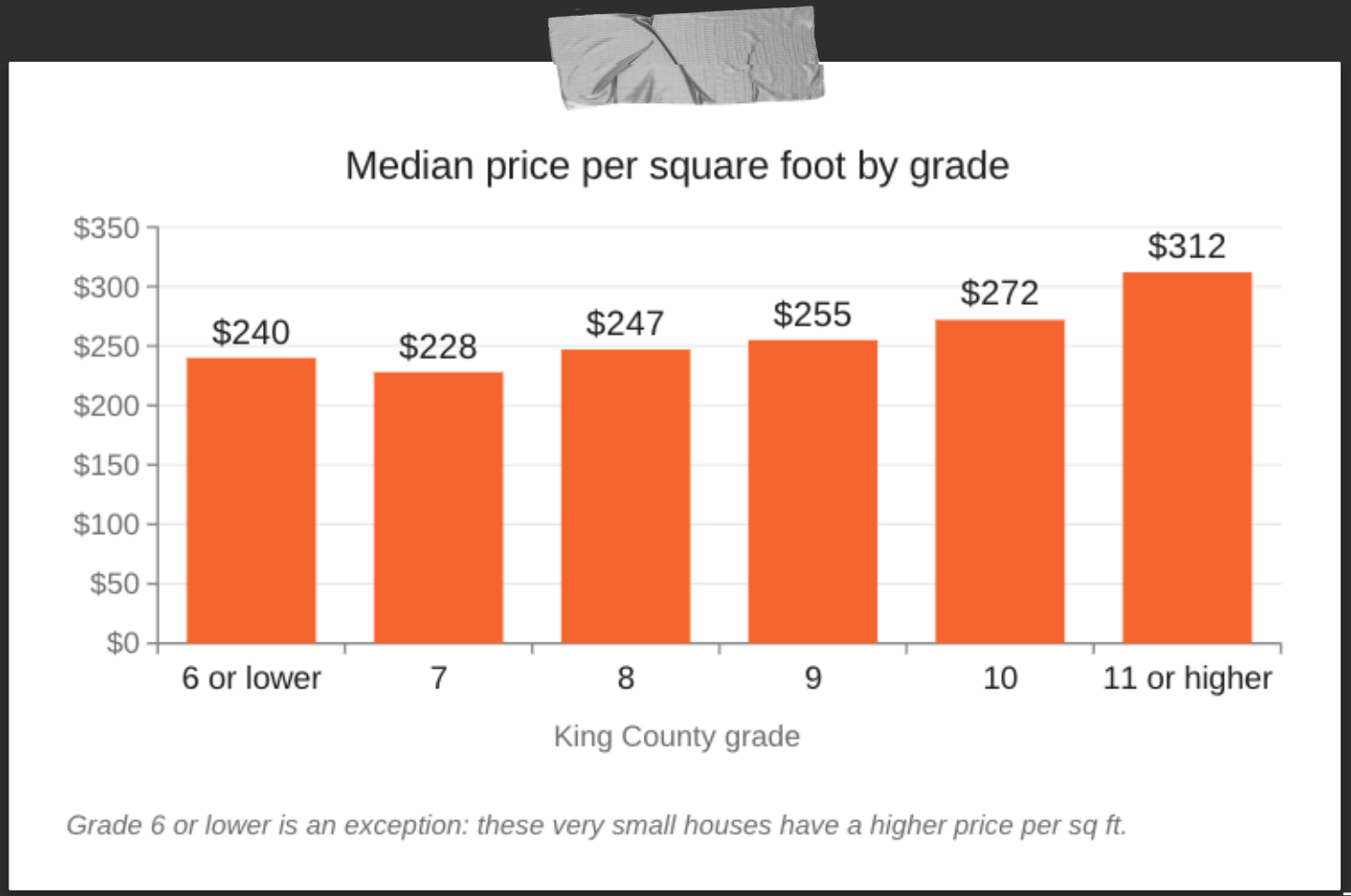
2. Better build quality costs more !

Higher grade, higher price per sq ft

From about $228 for grade 7 to $272 for grade 10: around 19% more for the same space.

The top grades jump

Grade 11 and higher costs about $312 per sq ft, around 37% more than grade 7.



Grade 6 or lower is an exception: these very small houses have a higher price per sq ft.

3. Location matters too !

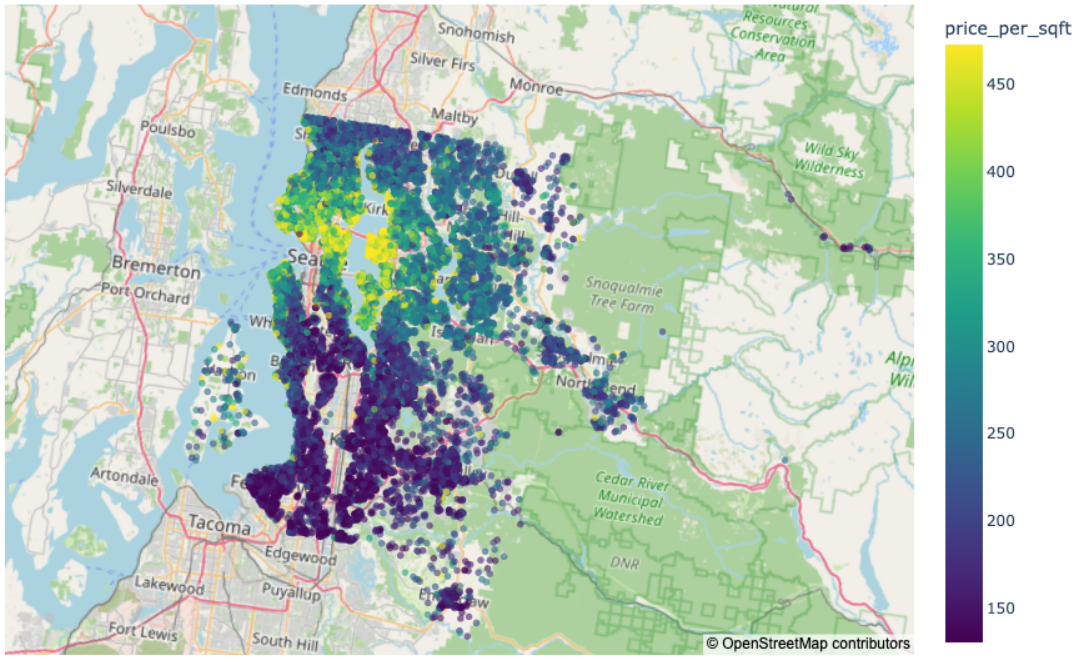
Most sales are in and around Seattle

Each dot is one sale, colored by price per sq ft.

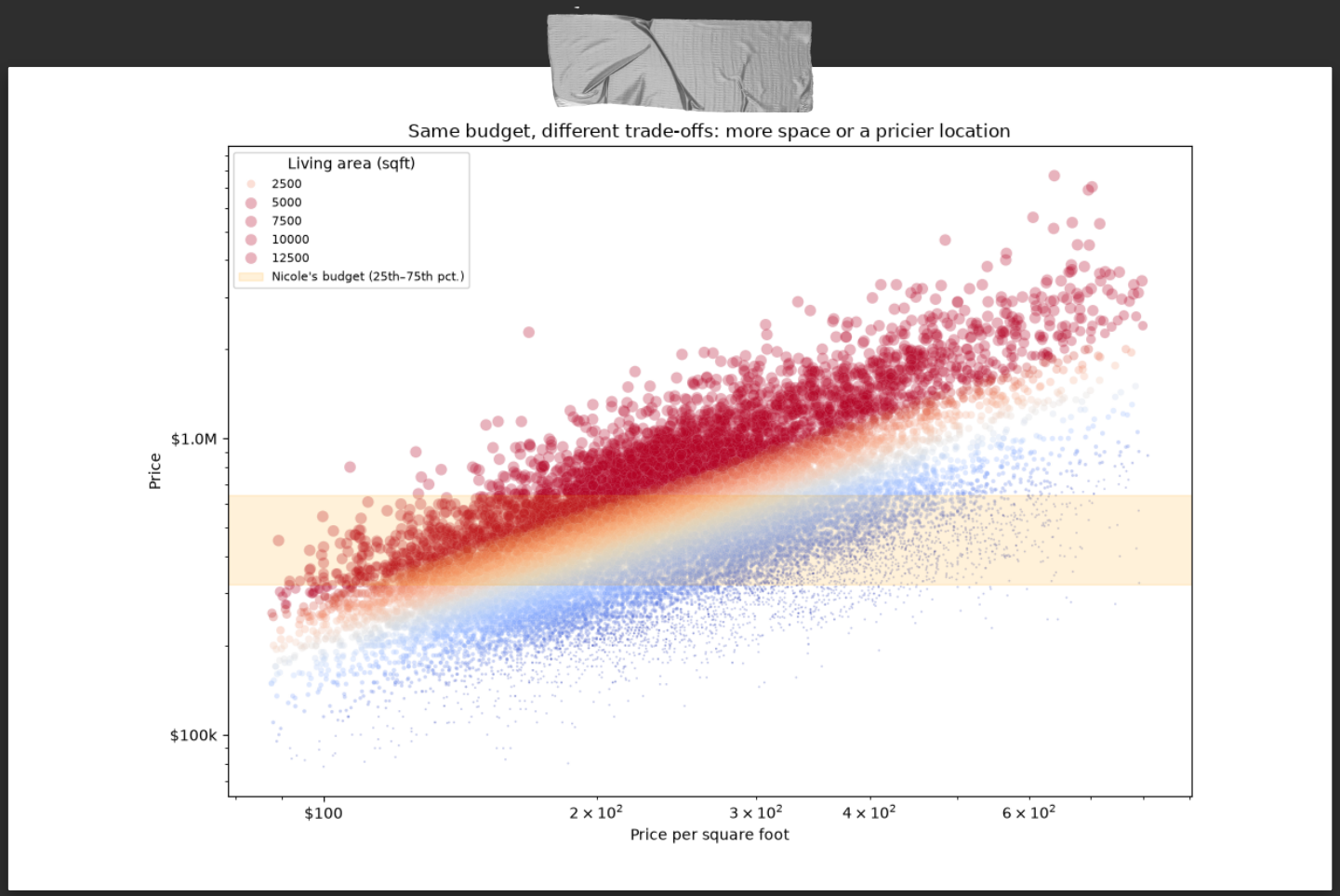
Central means expensive

Space costs most in and north of downtown Seattle and along Lake Washington (yellow), least in the south and far east (purple).

Open the interactive map



Prices and Nicole's budget



Nicole's budget

$322,000 – $645,000, the middle 50% of all prices (orange band).

Same budget, different trade-offs

Within the band, she can buy a large house that is cheap per sq ft (left in the plot), or a small house but more expensive price per sq ft (right).

Hypotheses

Hypo 1: Where?

The closer a house is to downtown Seattle, the higher its price. Mid-priced houses near the center will therefore be smaller or have a lower grade.

Hypo 2: When?

Houses sold in winter are cheaper than houses sold in spring and summer, so the buyer gets more for her money by buying in winter.

Hypo 3: What ?

Older houses near the center are cheaper than newer houses of similar size, so the buyer can afford a central home by choosing an older one.

Hypo 1:

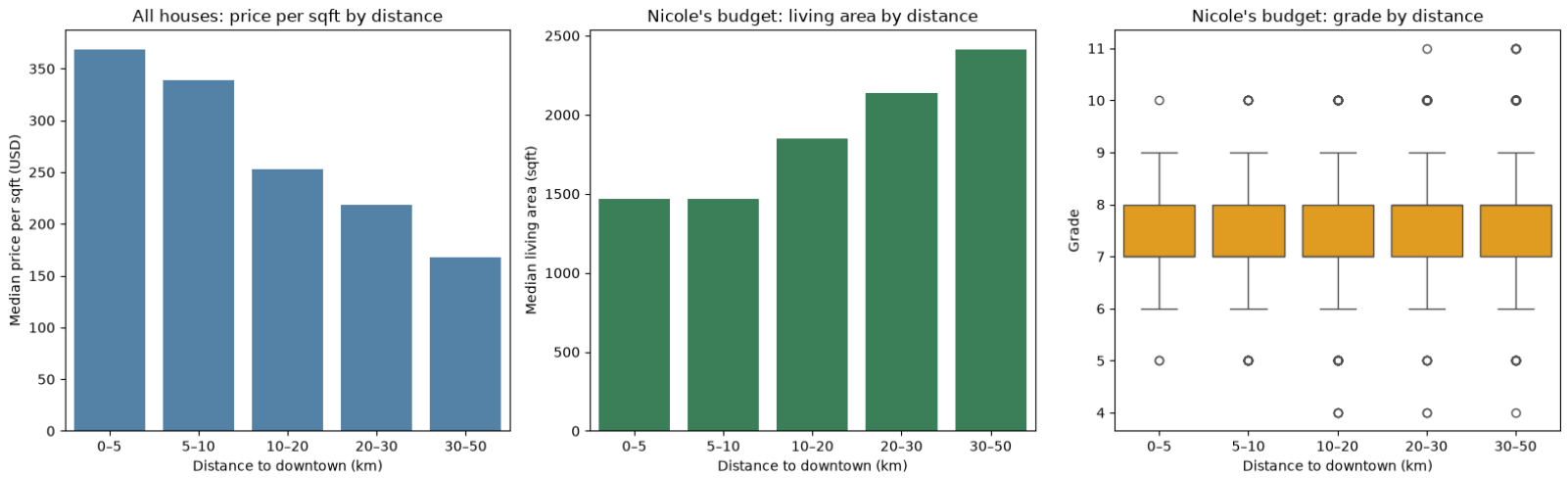
The closer to downtown, the higher the price. Mid-priced central houses are therefore smaller or lower grade.

Where to look at?

- price per sq ft
- distance to downtown
- living area and grade

How to look at?

- median per distance band
- size and grade only for houses in her budget



Partly supported. Central space costs over twice as much per sq ft ($370 within 5 km vs. $168 at 30–50 km). In her budget, central houses are smaller, but not lower grade (mostly 7–8 at every distance).

Hypo 1: a closer look at her budget

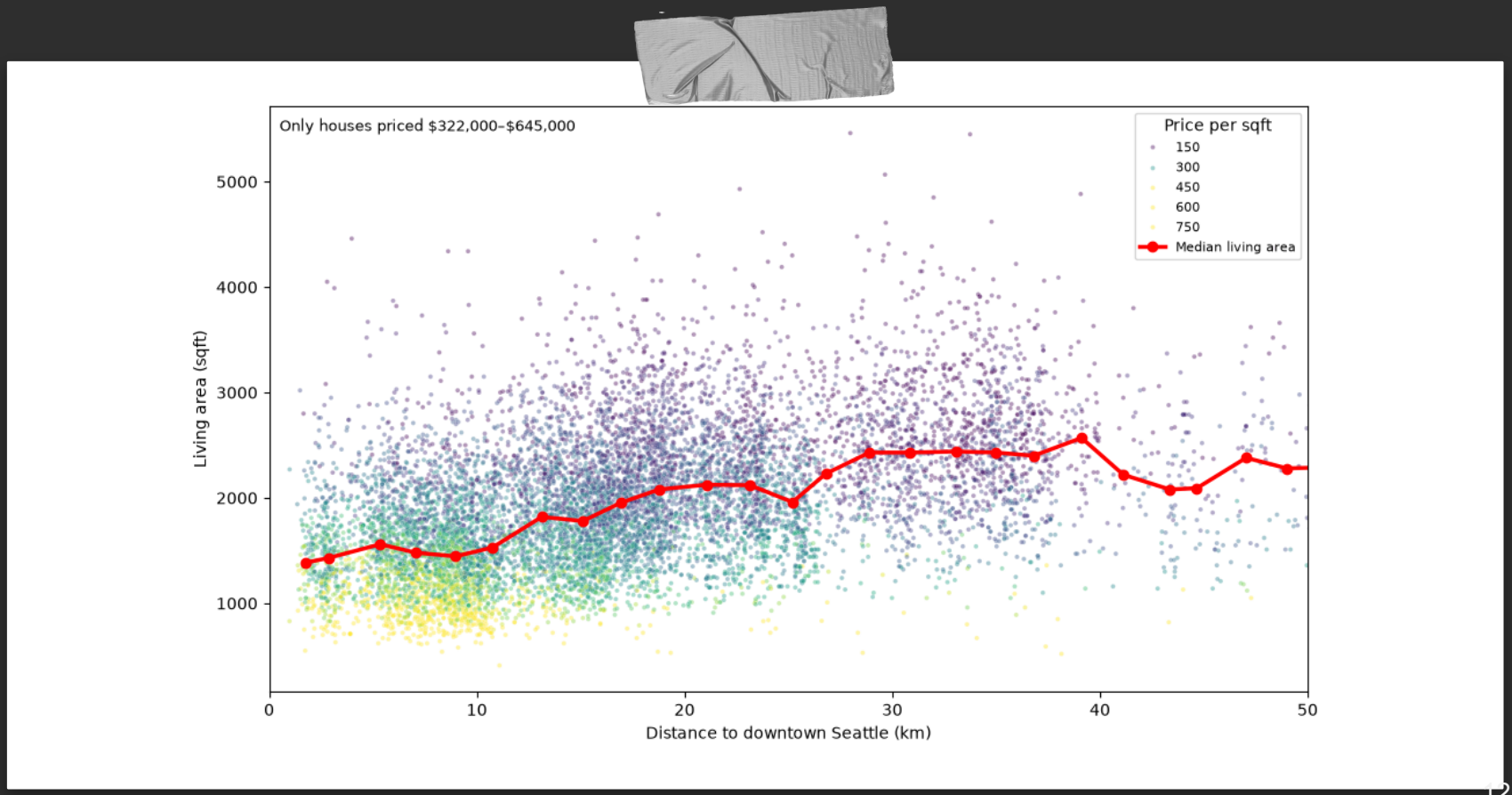
What does $322,000–$645,000 buy at each distance from downtown?

What do we see?

- each dot is one house in her budget
- red line: typical size at each distance
- yellow dots (pricey per sq ft) cluster near downtown

Result

- ≈1,470 sq ft within 10 km
- ≈2,400 sq ft at 30–50 km
- around 60% more space farther out



Hypo 2:

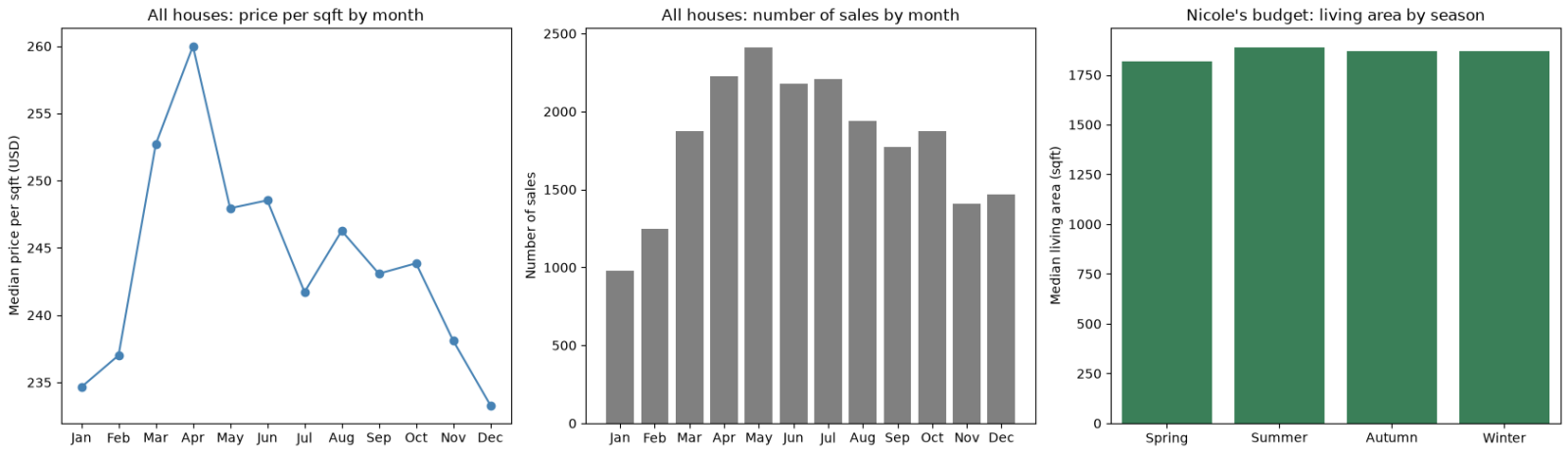
Houses sold in winter are cheaper than in spring and summer, so she gets more for her money in winter.

Where to look at?

- price per sq ft
- date of sale
- living area in her budget

How to look at?

- median per month and season
- county-wide first, then within 5 km of downtown



County-wide: yes. Prices peak in April (≈$260/sq ft) and are lowest in Dec–Jan (≈$233–235), about 10% lower. But far fewer houses sell in winter, and in her budget house size barely changes by season.

Hypo 2: within 5 km of downtown

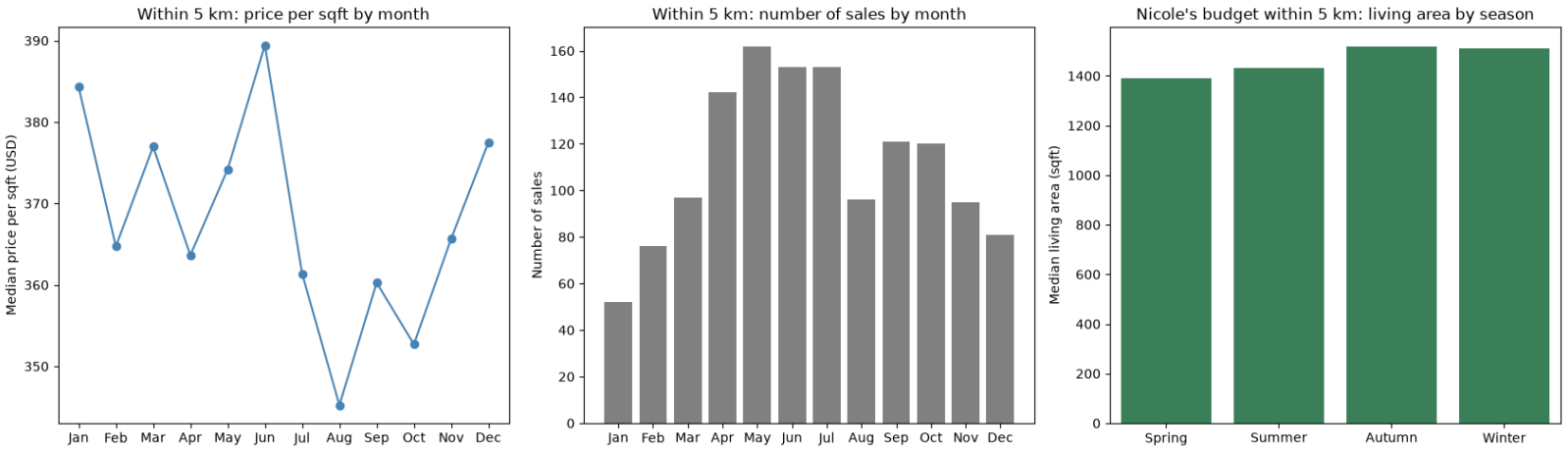
Does the seasonal pattern hold where Nicole wants to live?

What do we see?

- only 50–160 sales per month
- prices jump from month to month without a pattern
- slightly bigger houses in autumn and winter

Result

- no reliable seasonal effect in the center
- same grade in every season (≈7.5)



Partly supported. Winter is cheaper county-wide, but not reliably near downtown, where the differences are likely chance. Spring is the season to avoid.

Hypo 3:

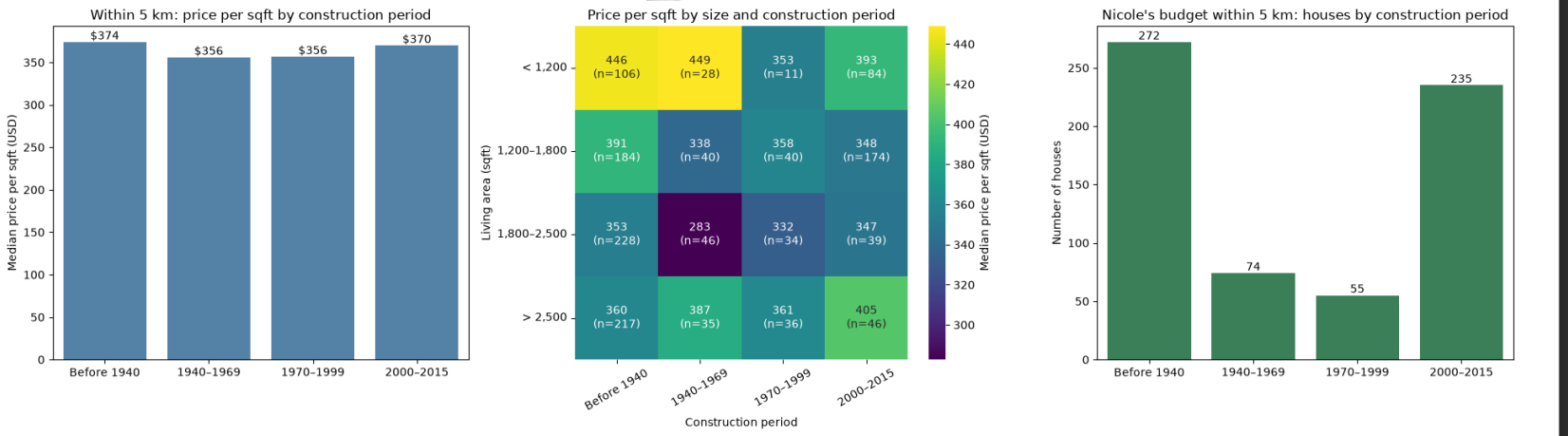
Older houses near the center are cheaper than newer houses of similar size.

Where to look at?

- price per sq ft
- year built
- living area

How to look at?

- houses within 5 km of downtown
- compare construction periods within the same size group
- count houses in her budget



Not supported. Pre-1940 houses cost the most per sq ft (≈$373), the newest ≈$369, and 1940–99 only about 5% less. No size group shows older = cheaper. Most central houses in her budget are either pre-1940 or post-2000.

Insights

Location matters most; Timing and age much less

1 Size and quality drive price

Grade (0.66) and living area (0.64) have the strongest link with price.

2 Central and north = pricier

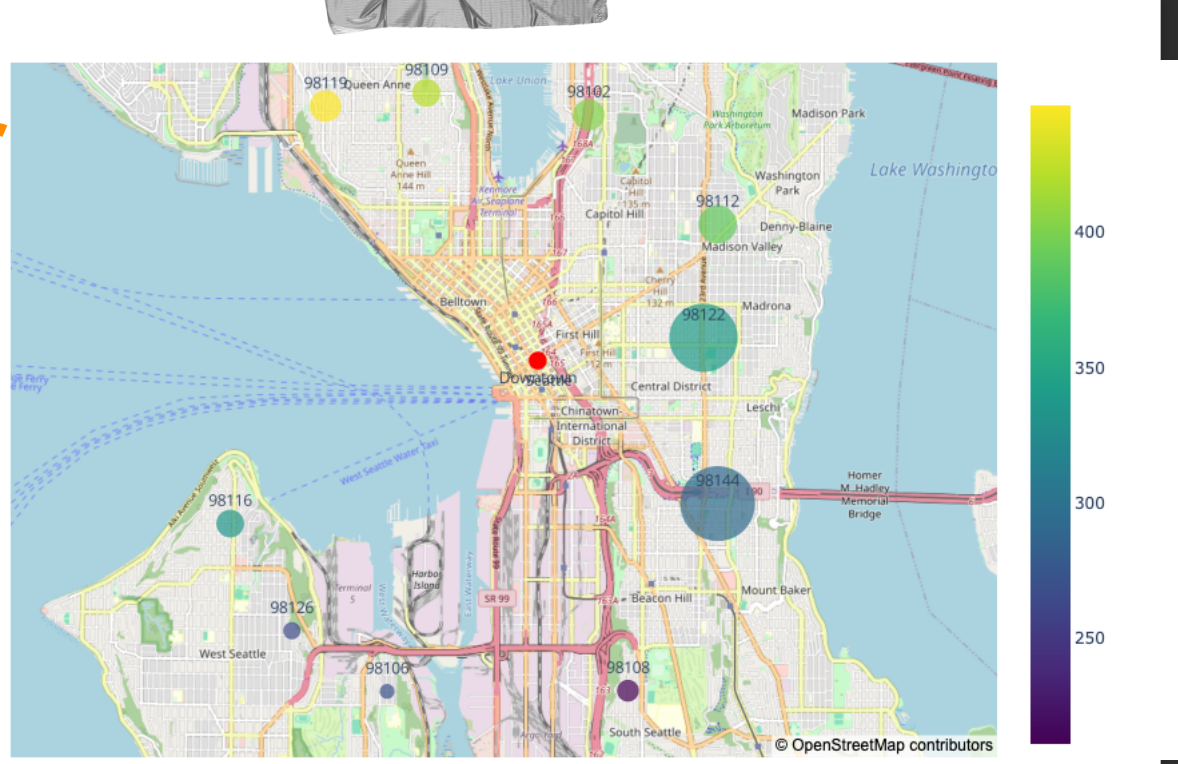
Within 5 km, the north (98119, 98109, 98102, 98112) costs most; 98122 and 98144 offer the best value.

3 Spring is the priciest season

County-wide, prices peak in April; December–January is about 10% cheaper.

4 Age barely matters

Old and new central houses cost about the same per sq ft.



Zip codes within 5 km, her budget. Circle size: houses sold. Color: price per sq ft (yellow = pricier).

My Recommendations

Where to search?

Focus on 98122 (Central District) and 98144 (Beacon Hill, Mount Baker): central, the most choice and a lower price per sq ft.

What to expect?

About 1,500 sq ft within 10 km of downtown, at the same build quality as further out.

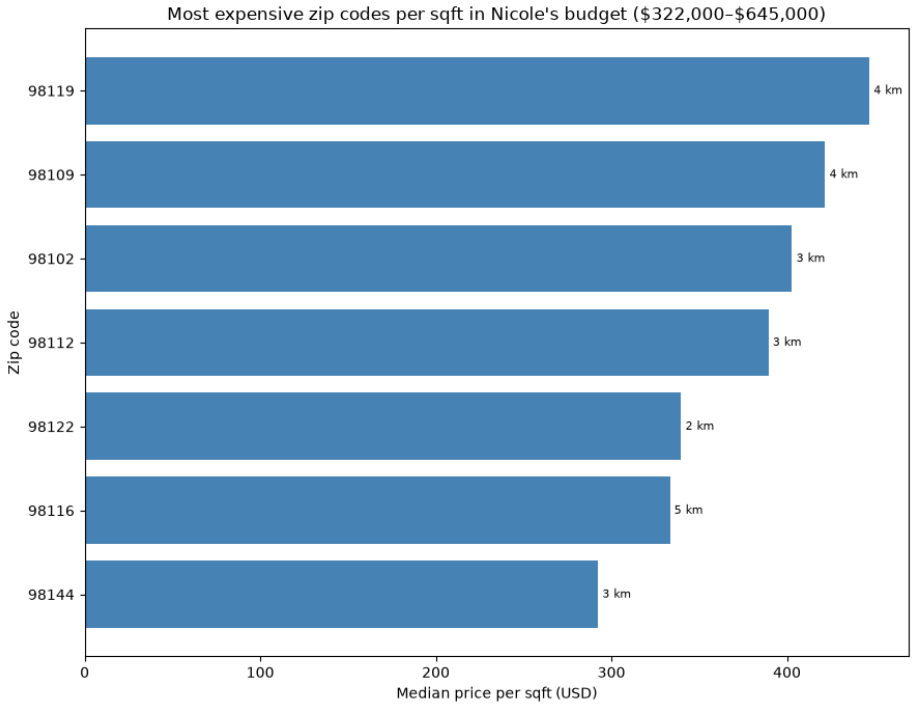
When to buy?

Skip spring (March–April). Late summer to autumn balances price and choice.

Old or new?

Choose by taste: older central houses aren't cheaper.

Price per sq ft in her budget, within 5 km



Thanks for listening.

